

Day and Night: Sky Shift

GROW • Science • 40 minutes

I can explain that Earth's rotation causes day and night and makes the Sun appear to move across our sky.

Prepare before pupils arrive

Setup: Mark one location clearly on the globe. Place a cool lamp at the same height and prepare four position cards.

Safety: Check the lamp stays cool, secure any cable, keep the room light high enough for safe movement and do not ask pupils to stare at the bulb.

Equipment: Cool LED lamp, globe or ball with a removable marker, four-position evidence sheet and pencil.

Safety: Use a cool LED light only. Never stare into a bright lamp. Keep cables still and let an adult move powered equipment.

Fallback: Use a fixed Sun card and a marked paper Earth. One pupil directs the paper Earth through four west-to-east quarter turns.

Access and participation

Offer reading aloud, pointing, signing, quiet thinking, a no-touch route and exact scribing where these support the learner. Keep the same subject decision. In shared work, let each learner commit before feedback; use a partner as card mover or checker, then swap. Avoid public speed rankings.

Source lesson curriculum link: Explain day and night and the apparent movement of the Sun.

40-minute teaching sequence

Stage / total time	Teaching move	Adult support / response
Arrival • 4 min	Last lesson: why can the Moon look different even though it stays spherical? Last week: state one way Earth moves in space.	Wait, regulate and retrieve through the lightest workable route. If recall is inaccessible, point to one retrieval sentence; do not teach today's answer.
Starter • 4 min	Think first. Point, say, sign or write your choice.	Protect curiosity: receive every first idea without judgement. Ask 'What makes you think that?' and preserve the prediction for later evidence.
I do • 5 min	rotate — turn axis — turning line apparent — seems to Fix a cool lamp as the Sun. Put one visible marker on a globe as our place.	Let the teacher/model carry the explanation; pupils watch, point or track. Use a visual cue only if access is the barrier, then fade it.

Stage / total time	Teaching move	Adult support / response
We do · 5 min	Commit to an answer. Show the evidence you used.	Scan every response mode. Do not infer understanding from handwriting or confidence. Secure: transfer. Mixed: contrast. Misconception: counterexample then a fresh question.
I do 2 · 4 min	Stand at the marked place in the model and describe the lamp's position from that viewpoint. As Earth rotates west to east, the observer turns with it.	Model one complete evidence-to-explanation sentence, then pause. Keep the science words; change density, modality or adult role before changing the goal.
We do 2 · 4 min	Match each observation at the marked place to its position in one rotation. Choose one evidence card.	Protect pupil operation and thinking; support handling only where needed. Ask what changed and what stayed the same before supplying a scientific word.
Independent · 10 min	Create a four-position explanation of day, night and the Sun's apparent movement for one marked place. Record: One saved or paper sequence with four positions, rotation arrows and a because sentence using apparent.	Protect independence. Accept equivalent evidence routes and scribe exact meaning. Prompt only as much as needed; fade as soon as the pupil continues.
Exit · 4 min	Explain why day and night happen and why the Sun appears to move across our sky. Point to, read or explain the evidence you made.	Genuinely receive one final response and name back the Science heard or seen. Choose one visible next move. The pupil writes what they said and what changed.

The timing belongs to the whole stage. Several PowerPoint slides may share it; do not restart that time on every slide. Full source-stage text is in the PowerPoint speaker notes and original HTML.

Answers, evidence and feedback

Hinge answer: Morning, because the place is entering the lit half.

Yes. The lamp remains fixed and the place rotates into the lit half, so morning begins there.

Misconception repair

Repaired. Earth rotates on its axis about once every 24 hours; the Sun's daily path is apparent.

Guided checkpoint

Yes. The lamp remains fixed and the place rotates into the lit half, so morning begins there. Keep the lamp fixed. Track only the marked place and ask whether it is entering or leaving the light.

Entering light

Sunrise. Sunrise: rotation carries the place into sunlight. Track the boundary. Moving from dark into light is the start of local daytime.

Facing the lamp

Middle of the day. Middle of the day: the place faces most directly towards the model Sun. Find the point on the lit half that faces the lamp most directly.

Leaving light

Sunset. Sunset: rotation carries the place out of sunlight. Moving from light into dark is the end of local daytime.

Facing away

Night. Night: this place faces away from the model Sun. Look for the position on the unlit half furthest from the lamp.

Model limitation

The lamp and globe are not to scale, the axis angle is simplified, and a quick turn represents about 24 hours.

Independent evidence

One saved or paper sequence with four positions, rotation arrows and a because sentence using apparent.

Supported: Order four picture cards and complete: When our place faces the Sun it is ___. When it faces away it is ___. Use the word bank: rotates, Sun, day, night, appears. An adult may scribe your exact words.

Standard: Draw and label four positions in one rotation, then explain day, night, sunrise and sunset. Use: Earth rotates, therefore our place..., so the Sun appears to....

Stretch: Explain the same four positions to someone who thinks the Sun orbits Earth each day. Include observer viewpoint, apparent movement and one honest model limitation.

Record the teaching response

What the learner actually showed	Next teaching action
Secure explanation with evidence	Reduce content prompting; offer another example or a limitation.
Partly correct / mixed	Point to the deciding evidence and invite a revision.
Access barrier	Change the reading, sensory or response format; keep the subject decision.
Missing source or observation	Keep the gap visible. Name the source or authorised check needed.

Safety and evidence boundary

Use the centre's authorised practical or fieldwork method and local risk assessment. Supplied sample data, fictional place models, card feedback and rehearsals are teaching material. They are not the learner's practical observations or proof of a qualification outcome. For portfolio use, check the exact registered unit/version, accepted evidence and centre process.

Sources and companion notes

Primary classroom source: SCI_G_W8A_Day_And_Night_Explore-1.html

The uploaded lesson is included unchanged. Text and teaching steps in the PowerPoint are editable; source diagrams are placed images. Word booklets are editable. Static PDFs cannot reproduce HTML controls; use the paper cards/tables or open the original HTML.

Verification scope: matched to the uploaded title, pathway, objective, stage sequence and 40-minute timing. Embedded workbook references are retained as source locators; this is not a new line-by-line audit of every scheme-of-work row or a centre assessment sign-off.